

FedSeq: A Hybrid Federated Learning Framework Based on Sequential In-Cluster Training

Zhikun Chen[✉], Daofeng Li[✉], Rui Ni[✉], *Member, IEEE*, Jinkang Zhu[✉], *Life Member, IEEE*, and Sihai Zhang[✉], *Senior Member, IEEE*

Abstract—Federated learning (FL) enables multiple devices to collaboratively accomplish a machine learning task by iteratively exchanging their model updates instead of raw data with a parameter server (PS), thus protecting users' data privacy. Nevertheless, the communication efficiency and the nonindependent and identically distributed (non-IID) data are remaining intractable challenges in FL. This article proposes a hybrid FL framework called FedSeq, based on user clustering and sequential in-cluster training, to improve the communication efficiency and the test accuracy, especially on non-IID data. The FedSeq first divides users into multiple clusters and selects a cluster head (CH) on behalf on each cluster to upload the model updates to the PS, thus drastically reducing the uplink communication overhead. Within each cluster, a sequential training method is designed, which enables the CHs' models witness more categories of training data and go through more meta-updates per training epoch, thus promoting the test accuracy. Moreover, we also provide the convergence analysis of FedSeq on IID data with Random user clustering, with specific experiments validating our theoretical analysis from a simulation perspective. In the experiments, we compare our FedSeq with other baselines, including FedSGD, FedAVG, FedProx, FedCluster, and HierFAVG, of which the results demonstrate that FedSeq outperforms other FL paradigms in terms of model accuracy and training efficiency. We also test FedSeq with Random and Low-Energy Adaptive Clustering Hierarchy clustering strategies to demonstrate the robustness of FedSeq to different clustering strategies. The FedSeq offers a potential solution to cope with non-IID data and reduce uplink communication overhead in FL.

Index Terms—Federated learning (FL), sequential in-cluster training, user clustering.

I. INTRODUCTION

FEDERATED learning (FL) is a burgeoning distributed machine learning framework due to its superiority in protecting users' privacy [1]. In FL, each user first downloads the initialized model from the central parameter server (PS) and then trains the model using its own data locally. After that, users upload their model updates to the PS, and finally, the PS performs an aggregation to build a new global model and sends it back to users. This process iterates until convergence. In this way, FL enables users to participate in a collaborative training while protecting their data privacy.

However, there are still intractable challenges in FL [2]. One is the excessive communication overhead produced during the FL training progress. To tackle this problem, one way is to reduce the size of model updates [3], [4], [5], [6]. In [3], structured updates and sketched updates are proposed. The former requires the model update to satisfy a predefined structure, while for the latter, a full model is learned and then compressed using subsampling, quantization, and random rotation, before uploading to the PS. In [4], an analog communication scheme called compressed analog distributed stochastic gradient descent (CA-DSGD) compresses the gradient to a finite number of bits and communicates these bits to the PS. In [5], a coreset-based FL explores the intrinsic redundant model parameters and dynamically removes the least important connections from the model. In [6], the FedKD shares a small mentee model between the server and clients instead of directly communicating the large models via a mutual knowledge distillation method. Aside from reducing the size of model updates by compression, the following studies reduce the communication frequency [7], [8]. In [7], a client-edge-cloud FL framework enables the edge server to aggregate its own users' model updates after several local iterations; the cloud server then aggregates the models of all the edge servers after every fixed number of edge model aggregation. In [8], a one-shot FL approach enables the server to learn a global model from a set of users in one single communication round. However, these works usually sacrifice the accuracy or the convergence rate of the model to reduce the communication overhead, which hinders their application scenarios. Another intractable problem in FL is the nonindependent and identically distributed (non-IID) data challenge, which refers to the inherent heterogeneity of different users' local data, including size imbalance and class distribution imbalance. Several techniques are proposed to address this challenge, including data augmentation [9], [10], multitask learning (MTL) [11], [12], transfer learning [13], [14], and so on. In [9], the Astraea addresses the class imbalance problem by augmenting training data according to the global data distribution. A large independent and identically distributed

Manuscript received 10 June 2022; revised 17 November 2022 and 11 January 2023; accepted 6 February 2023. Date of publication 24 February 2023; date of current version 30 August 2023. This work was supported in part by the National Key R&D Program of China under Grant 2022YFB2902302, in part by the Anhui Provincial Natural Science Foundation under Grant 2208085MF159, and in part by Fundamental Research Funds for the Central Universities under Grant KY2100000115. (*Corresponding author: Sihai Zhang.*)

Zhikun Chen and Daofeng Li are with the Department of Electronic Engineering and Information Science, School of Information Science and Technology, University of Science and Technology of China, Hefei 230026, China (e-mail: zhikunch@mail.ustc.edu.cn; df007@mail.ustc.edu.cn).

Rui Ni is with the Wireless Technology Laboratory, 2012 Laboratory, Huawei Tech. Co. Ltd., Shenzhen 518129, China (e-mail: raney.nirui@huawei.com).

Jinkang Zhu is with the PCNSS Laboratory, School of Information Science and Technology, University of Science and Technology of China, Hefei 230026, China (e-mail: jkzhu@ustc.edu.cn).

Sihai Zhang is with the Key Laboratory of Wireless-Optical Communications, Chinese Academy of Sciences, School of Microelectronics, University of Science and Technology of China, Hefei 230026, China (e-mail: shzhang@ustc.edu.cn).

Digital Object Identifier 10.1109/JSYST.2023.3243694

(IID) dataset is also built in the PS by collecting data from a small number of users that agree to share their data [10]. This dataset is then used for generating model updates that will be aggregated with the model updates trained on users' non-IID data. In [11], an MTL-based method called MOCHA enables each user to learn a separate model and simultaneously attempts to fit these related models. However, the MOCHA is restricted to linear MTL formulations; therefore, the authors further break this limitation and propose a fully practical federated kernelized MTL [12]. In [13], a federated transfer learning (FTL) framework is proposed to cope with the heterogeneity in users' data distribution, where a model learned from the source-domain data is applied on another different but related target-domain data with high accuracy. In [14], an FTL framework with discrepancy-based weighted federated averaging is proposed to train a good global diagnosis model collaboratively without exposing the privacy sensitive data. However, these works also have their limitations. For example, data augmentation requires to collect training samples for building an auxiliary dataset, which violates the principle of privacy protection. The MTL is too much computational complicated, especially when it is adopted to deep neural networks.

In this article, we devise a hybrid FL framework, called FedSeq, based on user clustering and sequential in-cluster training method, to reduce the uplink communication overhead and improve the model accuracy, especially on non-IID data. In FedSeq, users are first divided into multiple clusters according to some specific rules. Within each cluster, device-to-device (D2D) communication links enable users directly communicate with their neighbors for exchanging model parameters. The training process of FedSeq consists of two main stages: in-cluster training and model aggregation. During the in-cluster training, parallel for each cluster, users within the same cluster collaboratively train its local model based on its own data and deliver its intermediate trained model to its neighbor user in a relay-like fashion. Through this designed sequential in-cluster training method, the model generated on the cluster head (CH) witnesses more categories of training data and goes through more meta-updates during each training epoch, which potentially improves the model accuracy. After finishing in-cluster training, only one CH within each cluster is selected to upload its model updates to the PS. The PS then aggregates these models to build a global model and sends the aggregated model back to CHs to continue their next-epoch training. In this way, the uplink communication overhead can be drastically reduced since much fewer users upload the models to the server.

The contributions can be summarized as follows.

- 1) A hybrid FL framework called FedSeq is proposed in this article. By dividing users into multiple clusters and selecting only CHs to upload the model updates, the uplink communication overhead is drastically reduced. We also design a sequential in-cluster training scheme to enable the CH model witness more categories of data and go through more meta-updates per training epoch, which promotes the model accuracy.
- 2) We basically provide the convergence analysis of FedSeq on IID data with Random clustering strategy. Specific experiments show how the relevant parameters affect the convergence of FedSeq and empirically support our analysis from a simulation perspective.
- 3) We compare the test accuracy and training efficiency of FedSeq with other typical baselines in the experiment.

The results demonstrate that FedSeq outperforms other FL paradigms in terms of model accuracy and training efficiency. We also test FedSeq with different clustering methods, including Random and Low-Energy Adaptive Clustering Hierarchy (LEACH) strategies (see Section V-A), to show its robustness to different clustering strategies.

The rest of this article is organized as follows. In Section II, we provide some relevant studies that also implement FL based on user clustering. In Section III, we devise our solution FedSeq to solve the objective of this article. Then, in Section IV, the convergence of FedSeq is theoretically analyzed on IID data with Random user clustering. The experiments are introduced and discussed in Section V. Finally, Section VI concludes this article.

II. RELATED WORK

In this section, we introduce some existing studies related to the idea of user clustering [15], [16], [17], [18], [19], [20], [21], [22], [23], [24].

In [15], a federated multitask learning (FMTL) framework called clustered FL (CFL) is presented to address the suboptimal results of FL due to local users' non-IID data. In CFL, users are divided into multiple groups with jointly trainable data distributions according to the cosine similarity of FL losses. To address the problem of heterogeneous data distribution and Byzantine attack, a statistical model is proposed in [16], which takes into consideration the cluster structure of users' data. This model applies a Lloyd algorithm with l_2 -distance to group users' data. In [17], a hierarchical clustering mechanism also groups users based on the similarity of their local updates to the global model. In [18], CFL is further considered under constraints on the communication overhead, referred to as communication-aware CFL. An algorithm based on multisource adaptation methods is, therefore, proposed for this scenario, which allows the server to balance between learning more accurate models for each cluster and strengthening its ability to aggregate these models into a global one. In [19], the ClusterFL provides high model accuracy and low communication overhead for human activity recognition, by minimizing the empirical loss of learning models while capturing the intrinsic cluster structure among different users' data and dropping out the users that converge slower or have little correlation with other users in each cluster. In [20], an efficient framework called Iterative Federated Clustering Algorithm (IFCA) is designed for CFL, which alternately estimates the cluster identities of the users and optimizes model parameters for the user clusters via gradient descent. All the above studies should be, however, considered as an FMTL task rather than a traditional FL task, in which the users that belong to the same cluster jointly train a single model for their cluster. In [21], the FedCluster groups users into multiple clusters that perform FL cyclically in each learning round. This method can boost the convergence speed of FL since there are multiple cycles of model updates during each learning round. In [22], a hierarchical FL almost the same as the client-edge-cloud framework [7] aims to optimize the resource allocation among mobile users to reduce the communication latency in learning iterations. In [23], an efficient FL mechanism FedCH is proposed to accelerate FL in heterogeneous edge computing. According to the heterogeneous capacities, the FedCH arranges all users into multiple clusters, where users synchronously forward the model updates to a cluster header and the PS asynchronously aggregates models from cluster headers. Feng et al. [24] further developed the

TABLE I
COMPARISON OF RELATED WORK

Ref.	Objective	Idea	Setup	Theoretical analysis
[15]	improve Acc. on Non-IID data	group users by similarity of gradients	MTL	✓
[16]	improve Acc. on Non-IID data under Byzantine attack	use Lloyd algorithm to group users	MTL	✓
[17]	improve Acc. on Non-IID data	hierarchical clustering method	MTL	✗
[18]	improve [15] with communication constraint	multi-source adaptation method	MTL	✗
[19]	apply [15] to human activity recognition	—	MTL	✗
[20]	improve Acc. on Non-IID data	alternately group users and optimize user models	MTL	✓
[21]	improve Acc. on Non-IID data	group users and update models in a cluster-cyclic fashion	STL	✓
[22]	reduce the communication latency of FL	run partial aggregation on the edge before central server aggregation	STL	✗
[23]	reduce the training time and communication resources of [22]	hierarchical aggregation and asynchronous aggregation	STL	✓
[24]	consider [22] with user mobility	redesign the access mechanism, local update and global aggregation rule	STL	✓
This paper	improve Acc. on Non-IID data and reduce up-link communication overhead	user clustering, select CHs upload models and D2D enabled-sequential in-cluster training	STL	✓

MTL: multi-task learning; STL: single-task learning; ✓: support; ✗: not support.

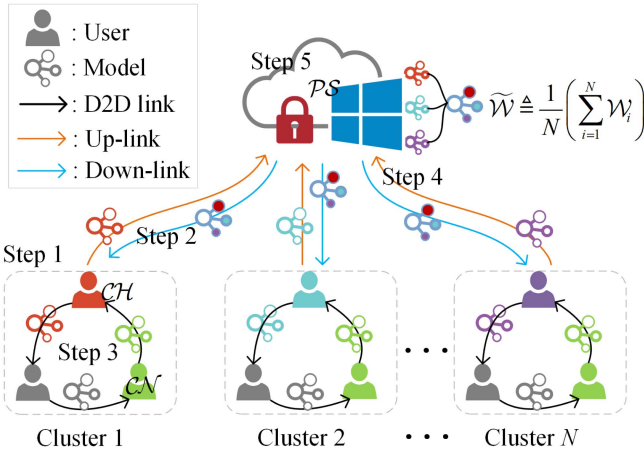


TABLE II
NOTATION

Symbol	Definition
N	number of clusters
T	number of training epochs
$\mathcal{K}_n$	number of users with the n -th cluster
$\bar{w}^t$	the averaged model at epoch t
w_n^t	the CH's model of the n -th cluster at epoch t
$w_{n,k}^t$	the model of the k -th user in the n -th cluster at epoch t
$\mathcal{G}_{n,k}^t$	gradients of $w_{n,k}^t$
$\mathcal{D}_{n,k}$	training set on the k -th user in the n -th cluster
$\mathcal{D}_n$	training set on the n -th cluster, $\mathcal{D}_n = \bigcup_{k \in \mathcal{K}_n} \mathcal{D}_{n,k}$
$\mathcal{D}$	the whole dataset, $\mathcal{D} = \bigcup_n \mathcal{D}_n$
$\xi_{n,k}$	training data sampled from $\mathcal{D}_{n,k}$
$\mathcal{F}$	loss function
η	learning rate
S_n^t	the in-cluster training order in cluster n at epoch t
T_{cm+cp}	training latency of each epoch
T_i^{cp}	computation latency of user i
T_{dc}^{cm}	communication latency between UE and cloud server
T_{dc}^{de}	communication latency between UE and edge server
T_{de}^{cm}	communication latency between two UEs
$\mathcal{E}_\alpha$	communication energy consumption to reach α accuracy

Fig. 1. Overview of the FedSeq framework.

client-edge-cloud FL into a mobility-aware cluster FL, where mobile users may roam across edge access points. Moreover, our preliminary work (Semi-FL) [25] also advocates to divide users into multiple clusters to the model accuracy.

In this article, we extend our previous Semi-FL [25] from the following aspects.

- 1) We employ more clustering methods, including Random and LEACH strategies, to enhance the robustness of our solution.
- 2) We provide the theoretical convergence analysis of our solution on IID data with Random user clustering.
- 3) We additionally measure the training efficiency and test accuracy of our solution by comparing it with other FL baselines including FL with no user clustering (FedSGD, FedAVG [1], and FedProx [26]) and FL with user clustering (FedCluster [21] and HierFAVG [7]).

Table I provides a more intuitive comparison of these studies.

III. METHODS

An overview of our proposed framework is indicated in Fig. 1, and Table II summarizes the notation in this article. Briefly,

all the users are first divided into multiple clusters according to some specific rules. At each training round, all the clusters parallelly perform its own in-cluster training, where each user updates and delivers its local model to its neighbor within the same cluster by our designed sequential in-cluster training method. After in-cluster training, each cluster selects a user equipment (UE) as the CH and uploads only the CH's model to the PS. The PS then averages these received models and sends the global averaged model back to the CHs. We formulate the objective of FedSeq as a distributed parallel optimization problem

$$\min_{w \in \mathbb{R}^d} f(w) \triangleq \frac{1}{N} \sum_{i=1}^N f_i(w) \quad (1)$$

where $f_i(w) = \mathbb{E}_{\zeta_i \sim \mathcal{D}_i} [\mathcal{F}_i(w; \zeta_i)]$ is the expectation form of loss function $\mathcal{F}_i$ over the training data $\mathcal{D}_i$ in the i th cluster, with

a model parameterized by $w \in \mathbb{R}^d$. In the following part, we describe this FedSeq framework in more detail.

A. FL With Sequential In-Cluster Training

As shown in Fig. 1, a total of five main steps constitute the training process of FedSeq.

Step 1 (User clustering): Users are divided into multiple clusters by the central server according to some specific rules. For example, users belonging to the same cell in a mobile communication system or users connecting to the same access point in a Wi-Fi system could be divided into the same cluster. In our experiment, we use two different strategies, i.e., Random clustering and LEACH clustering, to test the feasibility of our solution (elaborated in Section V).

Step 2 (CHs download global model): For each cluster, a CH selected by the central server downloads the global model from the PS to start or continue the in-cluster training. To support a possible dynamic clustering strategy, the CH user in FedSeq does not require to be fixed during the whole training process; instead, a user can be selected as a CH according to its communication or computation capabilities.

Step 3 (In-cluster training): Parallel for each cluster, each user within the same cluster collaboratively trains its own model based on its local training data and delivers this intermediate local-trained model to its neighbor. (Two users are called neighbors if they can directly communicate with each other through the D2D link within the same cluster.) After the in-cluster training is completed, a model “witnessing” all the training data within the cluster is generated on the CH, designated for uploading to the PS. A sequential in-cluster training method will be formally elaborated in Section III-B.

Step 4 (CHs upload models): Only the CHs on behalf of each cluster upload the models to the PS for global model aggregation. In this way, the FedSeq drastically reduces the uplink communication overhead by directly cutting down the number of models needed to be uploaded to the server.

Step 5 (Model aggregation): After receiving the models uploaded from all CHs, the PS performs model aggregation to build a global model by arithmetically averaging the received model parameters. All these steps would be repeated until achieving the convergence.

B. Sequential In-Cluster Training

In this section, we design a sequential training method to implement the in-cluster training, where each user trains its own model based on its local training data and then delivers the model to its neighbor to continue the neighbors’ training in a relay-like fashion. At round t , suppose that there are $\mathcal{K}_n$ users in cluster n with a sequential order $\mathcal{S}_n^t := 1, 2, \dots, \mathcal{K}_n$; then, the in-cluster training process can be formally expressed as

$$\begin{cases} w_{n,k}^t = \tilde{w}^{t-1} - \eta \frac{\sum_{\xi_{n,k} \in \mathcal{D}_{n,k}} \nabla \mathcal{F}(\tilde{w}^{t-1}; \xi_{n,k})}{|\mathcal{D}_{n,k}|}, & (k = 1) \\ w_{n,k}^t = w_{n,k-1}^t - \eta \frac{\sum_{\xi_{n,k} \in \mathcal{D}_{n,k}} \nabla \mathcal{F}(w_{n,k-1}^t; \xi_{n,k})}{|\mathcal{D}_{n,k}|}, & (k = 2, 3, \dots, \mathcal{K}_n) \end{cases} \quad (2)$$

where $w_{n,k}^t$ denotes the model of user k in the n th cluster at round t , $\mathcal{F}$ is the local loss function, η is the learning rate, $\tilde{w}^t = \frac{1}{N} \sum_{n=1}^N w_n^t$ is the average model downloaded from the PS, and $|\mathcal{D}_{n,k}|$ is the number of training examples of user’s dataset $\mathcal{D}_{n,k}$. After this, a model w_n^t “witnessing” all training data in the n th

Algorithm 1: Training Process of FedSeq.

```

1 Run User Clustering;
2 Select CHs for each cluster;
3 Initialization:  $\tilde{w}^0$ ,  $\eta$ , local batch size, sequential
   training order  $\mathcal{S}_n^0 := 1, 2, \dots, \mathcal{K}_n$  for  $n$ -th cluster;
4 Main Procedure: // main procedure
5 for  $t = 1, 2, 3, \dots, T$ , each cluster parallel do
6   if  $\text{regrouping} == \text{True}$  then
7     Run User Clustering;
8     Select the  $\mathcal{K}_n$ -th user as the CH;
9     Update training order  $\mathcal{S}_n^t$ ;
10  else
11    Keep the clustering result unchanged;
12     $\mathcal{S}_n^t \leftarrow \mathcal{S}_n^{t-1}$ ;
13  end
14  Run In-Cluster Training;
15  Generate the CH model:  $w_n^t = w_{n,\mathcal{K}_n}^t$ ;
16  CHs upload models to the PS;
17  Global model aggregation:  $\tilde{w}^t = \frac{1}{N} \sum_{n=1}^N w_n^t$ ;
18  The PS sends back the global model to the CHs;
19 end
20 Output:  $\tilde{w}^T$ ;

21 User Clustering: // sub procedure 1
22 Divide  $|N|$  users into a total of  $N$  clusters, so that the
    $n$ -th cluster has  $\mathcal{K}_n$  users;
23  $N_n = \text{Cluster}(N, \mathcal{K}_n)$ ;

24 In-Cluster Training: // sub procedure 2
25 for  $k \in \mathcal{S}_n^t$  do
26   Update the models sequentially according to (2);
27 end

```

cluster is generated on the CH and uploaded to the PS for model aggregation, where

$$w_n^t = w_{n,\mathcal{K}_n}^t. \quad (3)$$

The overall training process of FedSeq is summarized in Algorithm 1.

C. Practical Applicability

Since our solution introduces two additional steps, including *User Clustering* and *In-Cluster Training*, a potential problem arises that FedSeq to some extent complicates the FL process. Here, we demonstrate its practical applicability.

First, it is feasible for FedSeq to run *User Clustering*. To date, there are many widely used clustering methods, e.g., k -means [27], DBSCAN [28], and some clustering-based protocol in wireless sensor network, e.g., LEACH [29]. Therefore, the central server in FedSeq can be in charge of *User Clustering* according to specific clustering methods. Moreover, the local users’ contextual information, e.g., the communication and computation ability, or even the characteristic of data distribution can also be collected beforehand by the server to improve *User Clustering*.

Besides, *In-Cluster Training* can be facilitated by D2D communication. Usually, a smart UE nowadays is able to establish a D2D link with another UE, which ensures the feasibility of *In-Cluster Training*. Practically, the sequential training order

within each cluster can be either decided by the central server or by the selected CHs. Moreover, because *In-Cluster Training* is executed in a sequential relay-like way, instead of pairwise connected within a cluster, each user except for the CH is only required to be connected with its neighbor, which also reduces the burden of concurrent connections between all the users and the central server.

It should also be noted that we design a Random clustering strategy in our experiment only for validating the robustness of FedSeq to different clustering strategies. However, a user clustering strategy considering users' location information, like LEACH, is more encouraged to ensure users establish D2D links with their neighbors. To sum up, we claim that the FedSeq is still practically feasible if a protocol is properly designed over the PS-CH and CH-user links.

IV. CONVERGENCE ANALYSIS

In this section, we provide a basic theoretical analysis of FedSeq on IID data. As formulated in (1), since the loss function is identical across different users over different clusters in this article (e.g., cross-entropy loss), we use an unsubscripted $\mathcal{F}$ to replace $\mathcal{F}_i$ hereafter. There are assumed $K_n = K$ users within each cluster in our following analysis. For simplicity, we denote a local user's gradient information with respect to its own dataset $\mathcal{D}_{n,k}$ at round t as $\mathcal{G}_{n,k}^t$

$$\mathcal{G}_{n,k}^t = \frac{\sum_{\xi_{n,k} \in \mathcal{D}_{n,k}} \nabla \mathcal{F}(w_{n,k-1}^t; \xi_{n,k})}{|\mathcal{D}_{n,k}|}. \quad (4)$$

By expressing that the CH downloads the global average model at the previous round as

$$w_{n,0}^t = \tilde{w}^{t-1} \quad (5)$$

we can rewrite (2) as

$$w_{n,k}^t = w_{n,k-1}^t - \eta \mathcal{G}_{n,k}^t \quad (k = 1, 2, 3, \dots, K). \quad (6)$$

Therefore, we have

$$w_{n,K}^t = \tilde{w}^{t-1} - \eta \sum_{k=1}^K \mathcal{G}_{n,k}^t. \quad (7)$$

Before analyzing the convergence of FedSeq, we further require the problem (1) satisfy the three following assumptions.

Assumption 1: The CH observes an unbiased stochastic gradient, which can be treated as a representative over all data within the cluster, such that

$$\mathbb{E}_{\xi_{n,k} \sim \mathcal{D}_{n,k}} [\nabla \mathcal{F}(w_{n,k-1}^t; \xi_{n,k})] = \nabla f_n(w_n^{t-1}). \quad (8)$$

Following the IID data, (8) is empirically satisfied when the CH is randomly selected in each round.

Assumption 2: The expectation form of loss function f_i is L -smooth such that

$$\begin{cases} f_i(w) \leq f_i(v) + \langle \nabla f_i(v), w - v \rangle + \frac{L}{2} \|w - v\|^2 \\ \|\nabla f_i(w) - \nabla f_i(v)\| \leq L \|w - v\| \end{cases}. \quad (9)$$

Assumption 3: There exist constants σ and Ω such that

$$\mathbb{E}_{\xi_{i,k} \sim \mathcal{D}_{i,k}} \|\nabla \mathcal{F}(w; \xi_{i,k}) - \nabla f_i(w)\|^2 \leq \sigma^2 \quad \forall w, \forall i, \forall k \quad (10)$$

$$\mathbb{E}_{\xi_{i,k} \sim \mathcal{D}_{i,k}} \|\nabla \mathcal{F}(w; \xi_{i,k})\|^2 \leq \Omega^2 \quad \forall w, \forall i, \forall k. \quad (11)$$

In our experiments, we clip the gradients norm to satisfy the above two assumptions. According to the model aggregation in

Algorithm 1, we have

$$\tilde{w}^t = \frac{1}{N} \sum_{n=1}^N w_{n,K}^t. \quad (12)$$

By substituting (7) into (12), we have

$$\tilde{w}^t = \tilde{w}^{t-1} - \eta \frac{1}{N} \sum_{n=1}^N \sum_{k=1}^K \mathcal{G}_{n,k}^t. \quad (13)$$

Lemma 1: Under the above assumptions, the following inequality is ensured:

$$\mathbb{E} [\|\tilde{w}^t - w_{n,K}^t\|^2] \leq 4\eta^2 K^2 \Omega^2 \quad \forall n, \forall t. \quad (14)$$

In the following theorem, we take the average expected squared gradient norm to characterize the convergence rate of our solution.

Theorem 1: Under the above assumptions and lemmas, if the learning rate satisfies $0 \leq \eta \leq \frac{1}{LK}$, then, for $T \geq 1$, we have

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T \mathbb{E} [\|\nabla f(\tilde{w}^{t-1})\|^2] &\leq \frac{2}{\eta KT} (f(\tilde{w}^0) - f^*) \\ &\quad + 4\eta^2 K^2 L^2 \Omega^2 + \frac{L\eta K}{N} \sigma^2 \end{aligned} \quad (15)$$

where f^* denotes the minimum value of problem (1).

The theorem is proved by first deriving an upper bound on the deviation between the global averaged model with the CH model, i.e., Lemma 1, and then utilizing both Lemma 1 and the L -smooth assumption to derive the upper bound on the average expected squared gradient norm of the solution. Detailed proof is deferred to the Appendix.

In (15), the right part of the inequality can be decomposed into two parts. The “variant-part” $\frac{2}{\eta KT} (f(\tilde{w}^0) - f^*)$ would tend toward 0 as T increases. The “constant-part” $4\eta^2 K^2 L^2 \Omega^2 + \frac{L\eta K}{N} \sigma^2$ bounds the convergence rate, which can be further decomposed into two terms, i.e., $4\eta^2 K^2 L^2 \Omega^2$ and $\frac{L\eta K}{N} \sigma^2$. The former is related to the upper bound on the gradient of the local loss function, which depends on the assumption following (11). The latter is related to the deviation between the gradient on the local loss function and the gradient on the expected form of local loss function, which is originated from the deviation between the local training dataset and the whole training dataset. From (15), it can be observed that several hyperparameters possibly influence the convergence rate. Specifically, a larger η , K , and a smaller N would bring about a greater average expected squared gradient norm, which may lead to a faster convergence rate. To validate this suspect, we design specific experiments in Section V.

V. NUMERICAL RESULTS

In this section, the experimental results are analyzed following the elaboration of the experimental setups.

A. Experimental Setup

1) *Dataset Configuration:* We use MNIST [30] and CIFAR-10 [31] to validate our solution. Both IID and non-IID cases are considered when partitioning the dataset: 1) IID: each user is randomly assigned training samples with a uniform distribution over ten classes; and 2) non-IID: each user is assigned $\rho(0 < \rho \leq 1)$ proportion of training examples sampled from one major

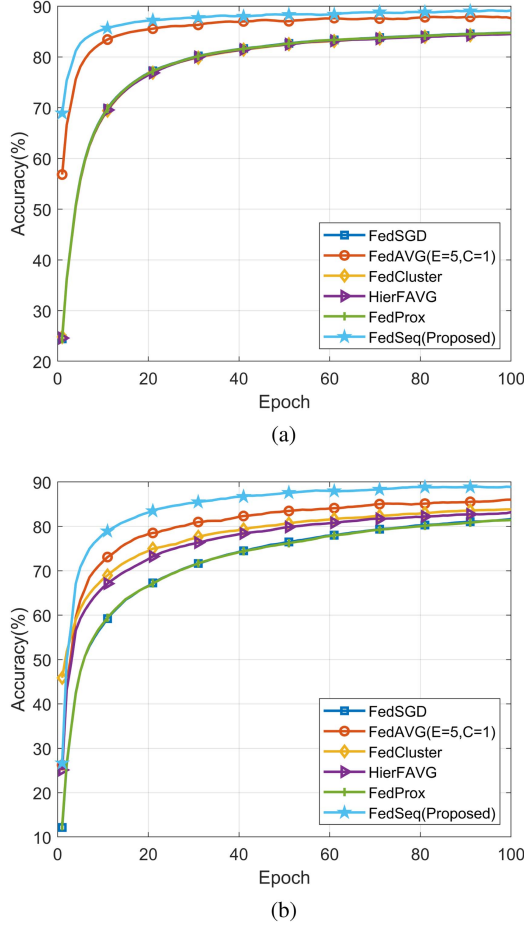


Fig. 2. Test accuracy on IID data. (a) MNIST IID. (b) CIFAR-10 IID.

class and $1 - \rho$ proportion sampled from other classes. Except for the result in Fig. 5, ρ is set as 0.8 by default.

2) *Clustering Strategies*: Two methods involving static and dynamic clustering strategies are designed in the experiment.

- Random-1: every ten users randomly sampled without replacement to form a cluster.
- Random-2: users are first divided into multiple clusters following Random-1 and additionally reclustered per ten training epochs.
- LEACH-1: users are randomly specified with coordinate (x_i, y_i) and then run the LEACH, a protocol utilizing users' coordinates to set up clusters in wireless sensors network [29].
- LEACH-2: users are first clustered following LEACH-1 and additionally respecified positions and reclustered per ten training epochs.

3) *Machine Learning Models*: To enhance the feasibility of our solution, we use a traditional support vector machine (SVM) [32] for MNIST classification and a very deep neural network VGG [31] for CIFAR-10 classification.

4) *Communication and Computation Models*: The communication and computation models are introduced for time efficiency and energy efficiency comparison in Section V-D.

Following [7], the uplink channel of UE to the edge is modeled as $B = 1$ MHz bandwidth with a gain $h = 10^{-8}$. The UE's transmitter power p is 0.5 W, and the noise power ϵ is 10^{-10} W.

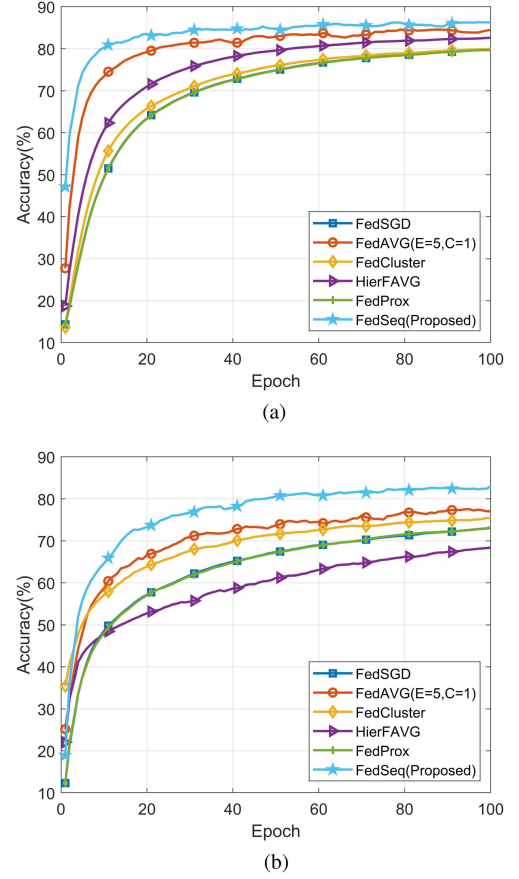


Fig. 3. Test accuracy on non-IID data. (a) MNIST non-IID. (b) CIFAR-10 non-IID.

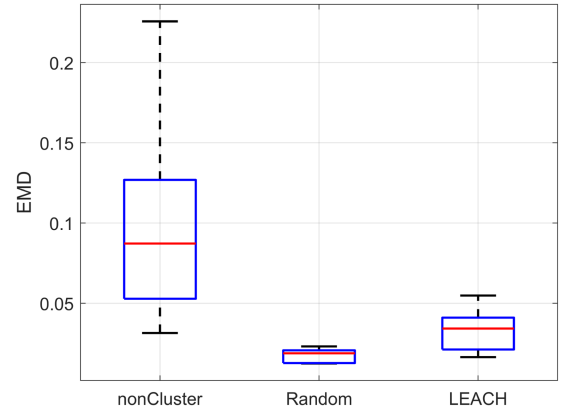


Fig. 4. EMD of different user's dataset or different cluster's dataset with the whole dataset in the non-IID scenario.

At each round, suppose that each UE uploads M bits of model updates; the communication latency from UE to the edge server is

$$T_{d2e}^{\text{cm}} = \frac{M}{B \log_2 \left(1 + \frac{hp}{\epsilon} \right)}. \quad (16)$$

Besides, we assume that the communication latency of UE to the cloud T_{d2c}^{cm} is $10 \times$ of that to the edge, while the local D2D

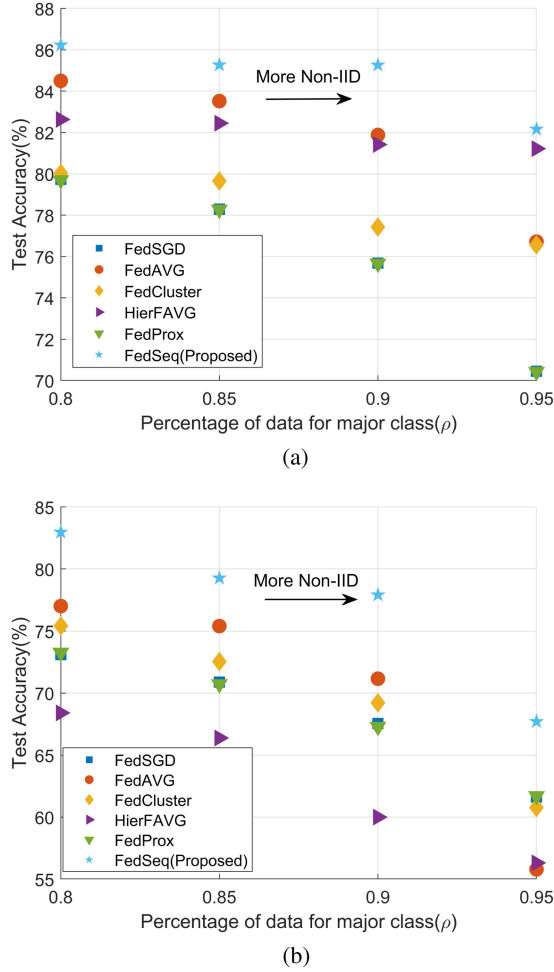


Fig. 5. Test accuracy on different degrees of non-IID. (a) MNIST. (b) CIFAR-10.

communication latency T_{d2d}^{cm} is $0.1 \times$ of that to the edge. We also ignore the downlink communication latency because of the adequate transmitting power of the server.

For the computation latency at an arbitrary round, we use the shifted exponential distribution [33] to characterize the computation latency of local model updating of user i :

$$\mathbb{P}[T_i^{cp} < t] = \begin{cases} 1 - e^{-\frac{\mu_i}{\tau d_i}(t - \alpha_i \tau d_i)}, & t \geq \alpha_i \tau d_i \\ 0, & \text{otherwise} \end{cases} \quad (17)$$

where $\alpha_i > 0$ and $\mu_i > 0$ are parameters indicating the maximum and fluctuation of computation capabilities, τ is the number of local iterations per epoch, and d_i is the batch size. In our experiments, we randomly generate $0 < \alpha_i < 1$ (ms/sample) and set $\mu_i = \frac{1}{\alpha_i}$ for each user to simulate the computation heterogeneity of different users. Moreover, we ignore the model aggregation latency on the edge and cloud server due to its powerful computation capability.

5) *Baselines*: Five baselines are used for our performance comparison: 1) FedAVG [1]: the federated averaging algorithm that samples a portion of users ($0 < C \leq 1$) with several local epochs ($E \geq 1$) to participate the model training at each communication round; 2) FedSGD: the simplest form of FedAVG with $C = 1$ and $E = 1$; 3) FedProx [26]: an FL method adding a proximal term to stabilize the training. The penalty constant is set

TABLE III
EXPERIMENTAL PARAMETERS

	MNIST (SVM)	CIFAR-10 (VGG11)
Local batch size	30	25
Learning rate	0.001	0.01
Decaying rate	0.995	
Number of users	50	
Training epochs	100	
Number of clusters	5	

as $\mu = 0.02$; 4) FedCluster [21]: a framework also divides users into multiple clusters, but these clusters perform FL cyclically in each learning round; and 5) HierFAVG [7]: a client-edge-cloud FL running multiple partial model aggregations on the edge server before executing a global model aggregation on the cloud server. The aggregation frequency constants are set as $\kappa_1 = 20$ and $\kappa_2 = 2$. It is important to note that solutions including FedAVG, FedProx, and our FedSeq all support hyper-parameter configuration with $0 < C < 1$ and $E > 1$. However, the HierFAVG does not support $E > 1$, while the FedCluster does not support $C < 1$. Therefore, for a fair comparison, except the FedAVG ($E = 5$), we set $E = 1$ for the other solutions and set $C = 1$ for all solutions.

The experiments are carried out on a Ubuntu 18.04 computer with four RTX 2080Ti GPUs. All the simulations are implemented by Python 3.8 and Pytorch 1.8.2 with CUDA 10.2, and we use MATLAB to visualize the result. Unless otherwise specified, some training parameters are set following Table III by default.

B. Test Accuracy Performance Comparison

Fig. 2 shows the test accuracy performance of different methods on IID data assignment with Random-1 clustering strategy. It can be observed that our FedSeq outperforms other baselines on both MNIST and CIFAR-10. For example, on CIFAR-10, our FedSeq finally achieves 88.89% accuracy after 100 training epochs, which attains around 5–7% improvement compared to the 83.83% accuracy for FedCluster, 83.21% for HierFAVG, 81.44% for FedProx, and 81.65% for FedSGD. Although running more local training epochs promotes the 81.65% accuracy of FedSGD to 86% accuracy for FedAVG, our FedSeq still outperforms FedAVG even with less local training epoch. This is mainly because the CHs' trained models in FedSeq have passed through more times of "meta-updates" by the designed sequential in-cluster training method.

As for the result of non-IID data in Fig. 3, our proposed FedSeq also outperforms other baselines in non-IID data. For example, in CIFAR-10, our FedSeq obtains 82.97% test accuracy, while the FedCluster obtains 75.42% accuracy, the HierFAVG obtains 68.4% accuracy, the FedProx achieves 73.26% accuracy, the FedAVG achieves 77.01% accuracy, and the FedSGD achieves 73.1% accuracy. Although all the methods have observed accuracy degradation in the non-IID case, an improvement of around 5–14% accuracy can still be obtained by our FedSeq compared to other baselines. The reason is that FedSeq reduces the non-IIDness between different clusters by user clustering. This suspect is validated in Fig. 4 by calculating the Earth mover's distance (EMD) [34] of labels between user's dataset with the whole dataset (depicted as nonCluster) and between each cluster's dataset with the whole dataset. As can be seen from Fig. 4, the EMDs of Random and LEACH are

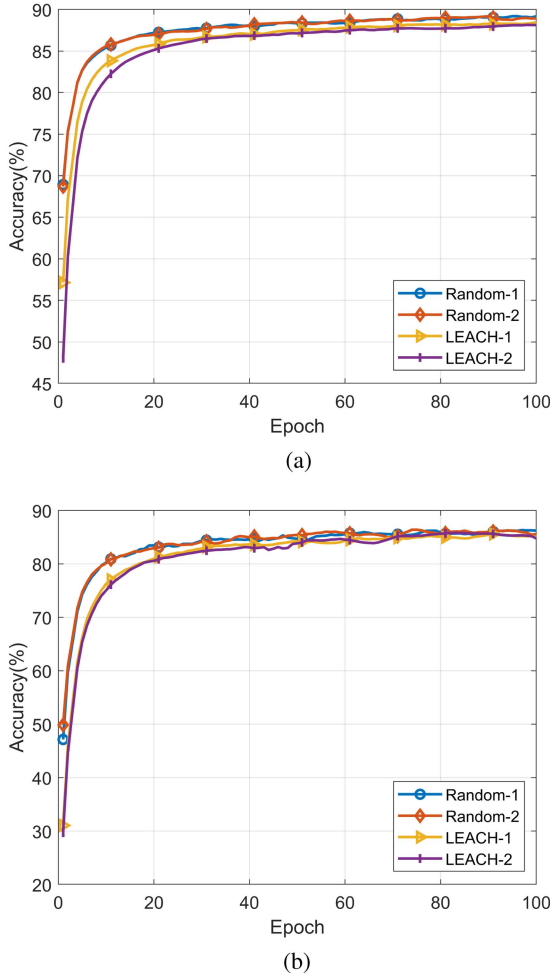


Fig. 6. Robustness of FedSeq to different clustering strategies. (a) MNIST IID. (b) MNIST non-IID.

much smaller than that of nonCluster, which indicates that a suitable clustering method is able to alleviate the non-IIDness of data distribution across different clusters. Besides, through the designed sequential in-cluster training method, the CHs' models have witnessed more categories of training data within the same cluster. In this way, the FedSeq has shown more competitive performance on non-IID data.

Moreover, we also compare the test accuracy of different methods with regard to different degrees of non-IID data in Fig. 5. Although the test accuracy of all the methods is degraded as the data distribution gets more non-IID, our proposed FedSeq still outperforms other baselines. For example, on CIFAR-10, FedSeq finally obtains 67.7% accuracy when $\rho = 0.95$, which outperforms other methods by around 5–10%.

To sum up, these results validate the feasibility and superiority of FedSeq on both IID and non-IID data with either traditional model SVM or deep neural network VGG.

C. Robustness to Different Clustering Strategies

To figure out the robustness of FedSeq to different clustering strategies, we test its performance with the clustering strategies elaborated in Section V-A2 and show the result in Fig. 6. Actually, we have tested the performance on both MNIST and

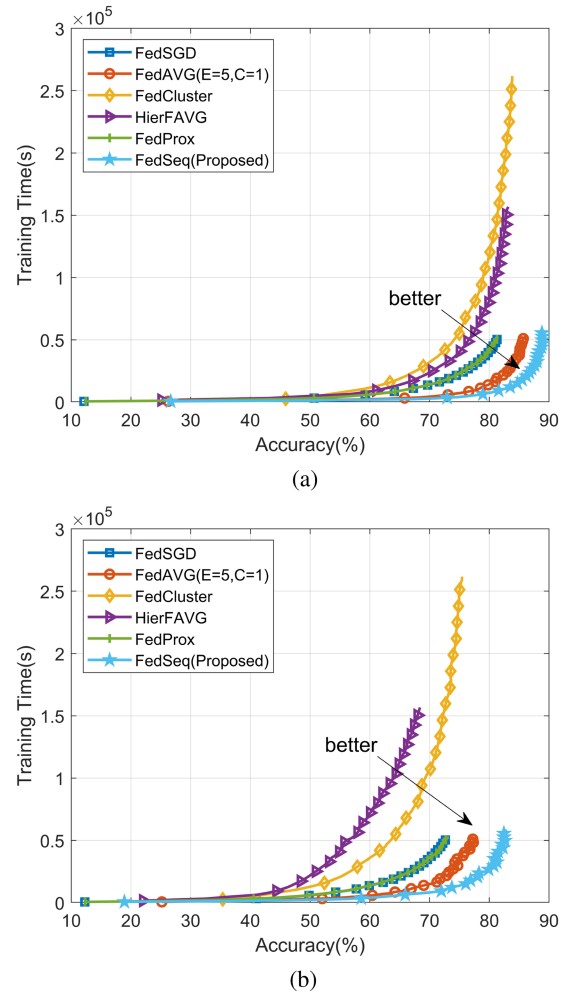


Fig. 7. Comparison of training time efficiency. (a) CIFAR-10 IID. (b) CIFAR-10 non-IID.

CIFAR-10, which have similar results, so we exemplify it with the result on MNIST here. From the result, we find that in both IID and non-IID cases, the Random-1 and Random-2 clustering strategies seem to converge faster than the LEACH strategies. The reason maybe that in Random strategies, all clusters have the same number of users, while in LEACH strategies, there are usually different number of users in each cluster. However, there is no significant final test accuracy between different clustering strategies, especially in the non-IID case. This demonstrates that our solution is scalable to different user clustering strategies.

Although our method shows good robustness to different clustering strategies, a method with a more even user clustering result like Random is suggested to obtain a faster convergence, while the LEACH is more favored in the wireless sensor network (WSN) scenario, where energy consumption is one of the most important considerations.

D. Performance of Time Efficiency and Energy Efficiency

In this section, we compare the training efficiency of different methods from two aspects, including time efficiency and energy efficiency, following the communication and computation model in Section V-A. The Random-1 clustering strategy is used

TABLE IV
COMPARISON OF ENERGY EFFICIENCY

Setups	Model (Bytes)	$\mathcal{E}$	FedSGD	FedAVG	FedProx	FedCluster	HierFAVG	FedSeq (Ours)
IID	MNIST-SVM (32513)	$\mathcal{E}_{80\%}$	344.04 J	68.8 J	355.51 J	378.44 J	1066.4 J	2.5 J
	CIFAR-VGG11 (36966521)	$\mathcal{E}_{75\%}$	586755 J	182546 J	586755 J	273819 J	1017068 J	9947 J
Non-IID	MNIST-SVM (32513)	$\mathcal{E}_{75\%}$	584.86 J	149.08 J	584.86 J	527.53 J	997.6 J	6.25 J
	CIFAR-VGG11 (36966521)	$\mathcal{E}_{65\%}$	534599 J	208624 J	534599 J	299897 J	2738260 J	15634.08 J

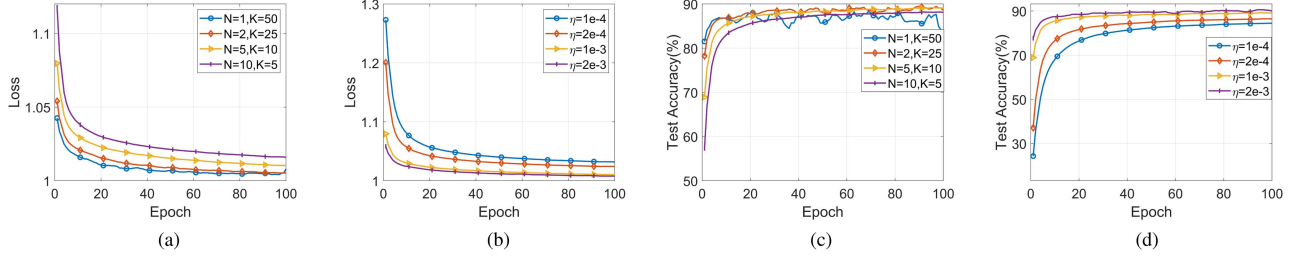


Fig. 8. Effect of K , N , and η on convergence rate and generalized test accuracy. (a) Loss versus N and K . (b) Loss versus η . (c) Accuracy versus N and K . (d) Accuracy versus η .

here. We first formulate the training time of different methods at an arbitrary round as (18), which comprises the uplink communication latency and the user's computation latency

$$T_{\text{cm+cp}} = \begin{cases} \max_{i \in \mathcal{N}} \left(\frac{\sum_{j=1}^{\kappa_1 \kappa_2} T_i^{\text{cp}}}{1000} + T_{\text{d2c}}^{\text{cm}} \right) & (\text{FedSGD \& FedProx}) \\ \max_{i \in \mathcal{N}} \left(\frac{\sum_{e=1}^E \sum_{j=1}^{\kappa_1 \kappa_2} T_i^{\text{cp}}}{1000} + T_{\text{d2c}}^{\text{cm}} \right) & (\text{FedAVG}) \\ \sum_{\mathcal{N}_i \subset \mathcal{N}} \left(\frac{\max_{i \in \mathcal{N}_i} \left(\sum_{j=1}^{\kappa_1 \kappa_2} T_i^{\text{cp}} \right)}{1000} + T_{\text{d2c}}^{\text{cm}} \right) & (\text{FedCluster}) \\ \max_{\mathcal{N}_i} \left(\frac{\sum_{j=1}^{\kappa_1 \kappa_2} T_i^{\text{cp}}}{1000} + \kappa_1 \cdot T_{\text{d2e}}^{\text{cm}} + T_{\text{d2c}}^{\text{cm}} \right) & (\text{HierFAVG}) \\ \max_{\mathcal{N}_i} \left(\frac{\sum_{i \in \mathcal{N}_i} \left(\sum_{j=1}^{\kappa_1 \kappa_2} T_i^{\text{cp}} \right)}{1000} + (|\mathcal{N}_i| - 1) \cdot T_{\text{d2d}}^{\text{cm}} + T_{\text{d2c}}^{\text{cm}} \right) & (\text{FedSeq}) \end{cases} \quad (18)$$

where $\mathcal{N}$ is a set of all user devices, $\mathcal{N}_i$ is subset of users in the i th cluster, N is the number of clusters, and $\kappa_1 \kappa_2$ is the total local iterations each user executes during each training epoch. In this experiment, since there are 50 users with the batch sizes 30 and 25 for MNIST and CIFAR-10, respectively, the local iterations per epoch here should be $\kappa_1 \kappa_2 = 40$ for each user. After this, we relate the time efficiency here to the training time required to achieve a predefined accuracy.

For the energy efficiency, note that we ignore the computation energy and consider only the total communication energy consumption produced by all the users during the training process. After this, we can calculate the total communication energy consumption per training epoch as

$$\mathcal{E} = \begin{cases} \sum_{i \in \mathcal{N}} (p \cdot T_{\text{d2c}}^{\text{cm}}) & (\text{FedSGD \& FedAVG \& FedProx \& FedCluster}) \\ \sum_{\mathcal{N}_i \subset \mathcal{N}} \left(\sum_{i \in \mathcal{N}_i} (p \cdot \kappa_1 \cdot T_{\text{d2e}}^{\text{cm}} + p \cdot T_{\text{d2c}}^{\text{cm}}) \right) & (\text{HierFAVG}) \\ \sum_{\mathcal{N}_i \subset \mathcal{N}} (p \cdot (|\mathcal{N}_i| - 1) \cdot T_{\text{d2d}}^{\text{cm}} + p \cdot T_{\text{d2c}}^{\text{cm}}) & (\text{FedSeq}) \end{cases} \quad (19)$$

We measure the energy efficiency by defining $\mathcal{E}_\alpha$ as the total communication energy consumption to achieve a predefined accuracy α and present the results of different methods in Table IV.

Since the results on both the datasets are similar, we exemplify the time efficiency by the training time consumption with regard

to the accuracy on CIFAR-10 in Fig. 7. It can be seen that the time efficiency of our FedSeq is superior to other baselines on both the datasets. More bluntly, the FedSeq takes the least training time to reach a preset threshold (e.g., 70%) among all these methods. The result indicates that although extra in-cluster communication latency $T_{\text{d2d}}^{\text{cm}}$ is introduced in the FedSeq, our solution can still have a higher time efficiency than other FL paradigms, because it can drastically reduce the number of training epochs through the sequential in-cluster training method. However, the result here although considers computation heterogeneity across different users but assumes no straggler within each cluster, other predominant strategies should be further considered in our future work when there exist stragglers during the training process.

For the energy efficiency in Table IV, we summarize the communication energy consumption of all the users to achieve α accuracy, $\mathcal{E}_\alpha$, where the result of our FedSeq is in bold. It is clear that the FedSeq has higher energy efficiency than that of other baselines, because the FedSeq consumes the least communication energy to achieve a predefined accuracy α . This is mainly because the FedSeq selects only CH users to communicate with the cloud and introduces less communication time $T_{\text{d2d}}^{\text{cm}}$ of D2D communication compared to the device-to-cloud communication time $T_{\text{d2c}}^{\text{cm}}$.

In a word, the FedSeq has an improved time efficiency and communication energy efficiency than other methods.

E. Convergence Versus η , K , and N

To validate our theoretical analysis in (15), we use SVM with different parameters K , N , and η on the MNIST dataset to observe the training loss and the generalized test accuracy in Fig. 8. The Random-1 clustering method is used here.

It can be seen from Fig. 8(a) that a larger K and correspondingly a smaller N lead to a faster decline of the training loss. In Fig. 8(b), a larger learning rate η also generally leads to a faster decline of training loss. This is consistent with our analysis in (15) that larger η and K and a smaller N would generate a larger average expected squared gradient norm, which, therefore, leads to a faster decline of the training loss. For the generalized accuracy in Fig. 8(c) and (d), a larger K corresponding to a smaller N also converges more quickly to higher accuracy in the early training epochs (e.g., epoch ≤ 10), which is consistent

with the result of training loss in Fig. 8(a). However, greater fluctuation for curves with $N = 1, K = 50$ and $N = 2, K = 25$ is observed in Fig. 8(c). This indicates that increasing K alone does not necessarily lead to good generalized accuracy because a greater K in FedSeq more likely leads to the problem similar to catastrophic forgetting. In Fig. 8(d), within a certain range, a larger learning rate is found to yield a faster convergence rate. This, however, does not always hold true because an excessive learning rate may cause unexpected fluctuations in the training process and makes it fail to converge. In summary, the above results support our mathematical analysis in (15) from a simulation perspective that, within a certain range, a larger K , a smaller N , and a larger η can make a larger dominant “constant-part” in (15), which indicates a larger average expected squared gradient norm and may lead to a faster convergence speed.

VI. CONCLUSION

FL is now becoming a ubiquitous distributed machine learning paradigm, benefiting from its advantage in privacy protection. In this article, we devise a hybrid FL called FedSeq, based on user clustering and sequential in-cluster training, to reduce the uplink communication overhead and improve the model accuracy. By user clustering, the FedSeq can group all the users into multiple clusters according to some specific rules. After this, only CH users are selected by the PS to upload their models for global model aggregation. In this way, the FedSeq directly cuts down the number of models needed to be uploaded, thus reducing the uplink communication overhead. Besides, through the sequential in-cluster training, the CH model is able to witness or eavesdrop more categories of training data within its cluster and go through more meta-updates per training round, which significantly promotes the model accuracy. Moreover, we theoretically analyze the convergence of our FedSeq on IID data with the Random user clustering strategy. Experimental results confirm the feasibility of our solution when applying different clustering strategies and indicate the advantage of FedSeq in test accuracy and training efficiency by comparing it with other existing FL paradigms.

One potential future work is to investigate the tradeoff between the convergence rate and the catastrophic forgetting [see Fig. 8(c)] during the sequential in-cluster training, design a more efficient clustering scheme through reinforcement learning, or automatically design a lightweight neural network model by neural architecture search.

APPENDIX

In this appendix, we provide an upper bound on the expected average squared gradient norms, which serves as a metric to measure the convergence rate. The local epoch is set as $E = 1$, and the Random-1 user clustering strategy is applied in the following proof such that each cluster contains K users.

Proof of Lemma 1:

Proof: Fix $t \geq 1$; for any two consecutive epochs t and $t - 1$, we have

$$w_n^t = w_{n,K}^t = \tilde{w}^{t-1} - \eta \sum_{k=1}^K \mathcal{G}_{n,k}^t. \quad (20)$$

From (13), $\tilde{w}^t = \tilde{w}^{t-1} - \eta \frac{1}{N} \sum_{n=1}^N \sum_{k=1}^K \mathcal{G}_{n,k}^t$, we have (21). Lemma 1 is therefore proved. ■

$$\begin{aligned} \mathbb{E} [\|\tilde{w}^t - w_n^t\|^2] &= \mathbb{E} \left[\left\| \eta \sum_{k=1}^K \frac{1}{N} \sum_{n=1}^N \mathcal{G}_{n,k}^t - \eta \sum_{k=1}^K \mathcal{G}_{n,k}^t \right\|^2 \right] \\ &= \eta^2 \mathbb{E} \left[\left\| \sum_{k=1}^K \frac{1}{N} \sum_{n=1}^N \mathcal{G}_{n,k}^t - \sum_{k=1}^K \mathcal{G}_{n,k}^t \right\|^2 \right] \\ &\leq 2\eta^2 \mathbb{E} \left[\left\| \sum_{k=1}^K \frac{1}{N} \sum_{n=1}^N \mathcal{G}_{n,k}^t \right\|^2 + \left\| \sum_{k=1}^K \mathcal{G}_{n,k}^t \right\|^2 \right] \\ &\leq 2\eta^2 K \mathbb{E} \left[\sum_{k=1}^K \left\| \frac{1}{N} \sum_{n=1}^N \mathcal{G}_{n,k}^t \right\|^2 + \sum_{k=1}^K \|\mathcal{G}_{n,k}^t\|^2 \right] \\ &\leq 2\eta^2 K \mathbb{E} \left[\sum_{k=1}^K \left(\frac{1}{N} \sum_{n=1}^N \|\mathcal{G}_{n,k}^t\|^2 \right) + \sum_{k=1}^K \|\mathcal{G}_{n,k}^t\|^2 \right] \\ &\leq 4\eta^2 K^2 \Omega^2. \end{aligned} \quad (21)$$

Proof of Theorem 1:

Proof: For $t \geq 1$, since f satisfies L -smooth condition, we have

$$\begin{aligned} \mathbb{E} [f(\tilde{w}^t)] &\leq \mathbb{E} [f(\tilde{w}^{t-1})] + \mathbb{E} [\langle \nabla f(\tilde{w}^{t-1}), \tilde{w}^t - \tilde{w}^{t-1} \rangle] \\ &\quad + \frac{L}{2} \mathbb{E} [\|\tilde{w}^t - \tilde{w}^{t-1}\|^2]. \end{aligned} \quad (22)$$

Besides, we can bound the l_2 norm of any two consecutive average model by (23)

$$\begin{aligned} \mathbb{E} [\|\tilde{w}^t - \tilde{w}^{t-1}\|^2] &= \eta^2 \mathbb{E} \left[\left\| \frac{1}{N} \sum_{n=1}^N \sum_{k=1}^K \mathcal{G}_{n,k}^t \right\|^2 \right] \\ &\stackrel{(a)}{=} \eta^2 \mathbb{E} \left[\left\| \frac{1}{N} \sum_{n=1}^N \left(\sum_{k=1}^K \mathcal{G}_{n,k}^t - K \nabla f_n(w_n^{t-1}) \right) \right\|^2 \right] \\ &\quad + \eta^2 \mathbb{E} \left[\left\| \frac{1}{N} \sum_{n=1}^N K \nabla f_n(w_n^{t-1}) \right\|^2 \right] \\ &\stackrel{(b)}{=} \eta^2 \frac{1}{N^2} \sum_{n=1}^N \mathbb{E} \left[\left\| \sum_{k=1}^K (\mathcal{G}_{n,k}^t - \nabla f_n(w_n^{t-1})) \right\|^2 \right] \\ &\quad + \eta^2 K^2 \mathbb{E} \left[\left\| \frac{1}{N} \sum_{n=1}^N \nabla f_n(w_n^{t-1}) \right\|^2 \right] \\ &\stackrel{(c)}{\leq} \eta^2 \frac{K}{N^2} \sum_{n=1}^N \mathbb{E} \left[\sum_{k=1}^K \|\mathcal{G}_{n,k}^t - \nabla f_n(w_n^{t-1})\|^2 \right] \\ &\quad + \eta^2 K^2 \mathbb{E} \left[\left\| \frac{1}{N} \sum_{n=1}^N \nabla f_n(w_n^{t-1}) \right\|^2 \right] \end{aligned}$$

$$\stackrel{(d)}{\leq} \frac{K^2}{N} \eta^2 \sigma^2 + \eta^2 K^2 \mathbb{E} \left[\left\| \frac{1}{N} \sum_{n=1}^N \nabla f_n(w_n^{t-1}) \right\|^2 \right] \quad (23)$$

where (a) follows by $\mathbb{E}[\|\mathbf{Z}\|^2] = \mathbb{E}[\|\mathbf{Z} - \mathbb{E}[\mathbf{Z}]\|^2] + \|\mathbb{E}[\mathbf{Z}]\|^2$ for any vector $\mathbf{Z}$, (b) follows by (8), (c) follows by applying $\|\sum_{i=1}^n \mathbf{z}_i\|^2 \leq n \sum_{i=1}^n \|\mathbf{z}_i\|^2$ for any vector $\mathbf{z}_i$, and (d) follows (10).

By the iterated law of expectations and the basic equality $\langle \mathbf{z}_1, \mathbf{z}_2 \rangle = \frac{1}{2}(\|\mathbf{z}_1\|^2 + \|\mathbf{z}_2\|^2 - \|\mathbf{z}_1 - \mathbf{z}_2\|^2)$, we further get (24).

$$\begin{aligned} & \mathbb{E}[\langle \nabla f(\tilde{w}^{t-1}), \tilde{w}^t - \tilde{w}^{t-1} \rangle] \\ &= -\eta \mathbb{E} \left[\left\langle \nabla f(\tilde{w}^{t-1}), \frac{1}{N} \sum_{n=1}^K \sum_{k=1}^K \mathcal{G}_{n,k}^t \right\rangle \right] \\ &= -\eta \mathbb{E} \left[\left\langle \nabla f(\tilde{w}^{t-1}), \frac{1}{N} \sum_{n=1}^N K \nabla f_n(w_n^{t-1}) \right\rangle \right] \\ &= -\eta K \mathbb{E} \left[\left\langle \nabla f(\tilde{w}^{t-1}), \frac{1}{N} \sum_{n=1}^N \nabla f_n(w_n^{t-1}) \right\rangle \right] \\ &= -\frac{\eta K}{2} \mathbb{E} \left[\|\nabla f(\tilde{w}^{t-1})\|^2 + \left\| \frac{1}{N} \sum_{n=1}^N \nabla f_n(w_n^{t-1}) \right\|^2 \right] \\ &\quad - \frac{\eta K}{2} \mathbb{E} \left[\left\| \nabla f(\tilde{w}^{t-1}) - \frac{1}{N} \sum_{n=1}^N \nabla f_n(w_n^{t-1}) \right\|^2 \right] \quad (24) \end{aligned}$$

Combining (23) and (24) with (22), we have (25)

$$\begin{aligned} & \mathbb{E}[f(\tilde{w}^t)] \leq \mathbb{E}[f(\tilde{w}^{t-1})] \\ &\quad - \frac{\eta K - \eta^2 L K^2}{2} \mathbb{E} \left[\left\| \frac{1}{N} \sum_{n=1}^N \nabla f_n(w_n^{t-1}) \right\|^2 \right] \\ &\quad - \frac{\eta K}{2} \mathbb{E}[\|\nabla f(\tilde{w}^{t-1})\|^2] + \frac{L}{2N} K^2 \eta^2 \sigma^2 \\ &\quad + \frac{\eta K}{2} \mathbb{E} \left[\left\| \nabla f(\tilde{w}^{t-1}) - \frac{1}{N} \sum_{n=1}^N \nabla f_n(w_n^{t-1}) \right\|^2 \right] \\ &\stackrel{(a)}{\leq} \mathbb{E}[f(\tilde{w}^{t-1})] - \frac{\eta K - \eta^2 L K^2}{2} \mathbb{E} \left[\left\| \frac{1}{N} \sum_{n=1}^N \nabla f_n(w_n^{t-1}) \right\|^2 \right] \\ &\quad - \frac{\eta K}{2} \mathbb{E}[\|\nabla f(\tilde{w}^{t-1})\|^2] + 2\eta^3 K^3 L^2 \Omega^2 + \frac{L K^2 \eta^2 \sigma^2}{2N} \\ &\stackrel{(b)}{\leq} \mathbb{E}[f(\tilde{w}^{t-1})] - \frac{\eta K}{2} \mathbb{E}[\|\nabla f(\tilde{w}^{t-1})\|^2] \\ &\quad + 2\eta^3 K^3 L^2 \Omega^2 + \frac{L K^2 \eta^2 \sigma^2}{2N} \quad (25) \end{aligned}$$

where (b) follows from $0 \leq \eta \leq \frac{1}{LK}$ and (a) follows (26)

$$\mathbb{E} \left[\left\| \nabla f(\tilde{w}^{t-1}) - \frac{1}{N} \sum_{n=1}^N \nabla f_n(w_n^{t-1}) \right\|^2 \right]$$

$$\begin{aligned} &= \mathbb{E} \left[\left\| \frac{1}{N} \sum_{n=1}^N \nabla f_n(\tilde{w}^{t-1}) - \frac{1}{N} \sum_{n=1}^N \nabla f_n(w_n^{t-1}) \right\|^2 \right] \\ &= \frac{1}{N^2} \mathbb{E} \left[\left\| \sum_{n=1}^N (\nabla f_n(\tilde{w}^{t-1}) - \nabla f_n(w_n^{t-1})) \right\|^2 \right] \\ &\leq \frac{1}{N} \mathbb{E} \left[\sum_{n=1}^N \|\nabla f_n(\tilde{w}^{t-1}) - \nabla f_n(w_n^{t-1})\|^2 \right] \\ &\leq L^2 \frac{1}{N} \sum_{n=1}^N \mathbb{E}[\|\tilde{w}^{t-1} - w_n^{t-1}\|^2] \leq 4L^2 \eta^2 K^2 \Omega^2. \quad (26) \end{aligned}$$

Dividing both sides of (25) by $\frac{\eta K}{2}$ and rearranging terms, we obtain

$$\begin{aligned} \mathbb{E}[\|\nabla f(\tilde{w}^{t-1})\|^2] &\leq \frac{2}{\eta K} (\mathbb{E}[f(\tilde{w}^{t-1})] - \mathbb{E}[f(\tilde{w}^t)]) \\ &\quad + 4\eta^2 K^2 L^2 \Omega^2 + \frac{L\eta K}{N} \sigma^2. \quad (27) \end{aligned}$$

Finally, we sum over $t \in \{1, \dots, T\}$ for (27) and divide both sides by T

$$\begin{aligned} & \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(\tilde{w}^{t-1})\|^2] \\ &\leq \frac{2}{\eta K T} (f(\tilde{w}^0) - \mathbb{E}[f(\tilde{w}^T)]) + 4\eta^2 K^2 L^2 \Omega^2 + \frac{L\eta K}{N} \sigma^2 \\ &\leq \frac{2}{\eta K T} (f(\tilde{w}^0) - f^*) + 4\eta^2 K^2 L^2 \Omega^2 + \frac{L\eta K}{N} \sigma^2 \quad (28) \end{aligned}$$

where the second inequality holds because f^* is the minimum value of problem (1). ■

REFERENCES

- [1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. 20th Int. Conf. Artif. Intell. Statist.*, 2017, vol. 54, pp. 1273–1282.
- [2] O. A. Wahab, A. Mourad, H. Otrok, and T. Taleb, "Federated machine learning: Survey, multi-level classification, desirable criteria and future directions in communication and networking systems," *IEEE Commun. Surv. Tuts.*, vol. 23, no. 2, pp. 1342–1397, Second Quarter 2021.
- [3] J. Konečný, H. B. McMahan, F. X. Yu, P. Richtárik, A. T. Suresh, and D. Bacon, "Federated learning: Strategies for improving communication efficiency," *CoRR*, vol. abs/1610.05492, 2016. [Online]. Available: <http://arxiv.org/abs/1610.05492>
- [4] M. M. Amiri and D. Gündüz, "Federated learning over wireless fading channels," *IEEE Trans. Wireless Commun.*, vol. 19, no. 5, pp. 3546–3557, May 2020.
- [5] K. Li and C. Xiao, "CBFL: A communication-efficient federated learning framework from data redundancy perspective," *IEEE Syst. J.*, vol. 16, no. 4, pp. 5572–5583, Dec. 2021.
- [6] C. Wu, F. Wu, L. Lyu, Y. Huang, and X. Xie, "Communication-efficient federated learning via knowledge distillation," *Nature Commun.*, vol. 13, no. 1, 2022, Art. no. 2032.
- [7] L. Liu, J. Zhang, S. Song, and K. B. Letaief, "Client-edge-cloud hierarchical federated learning," in *Proc. IEEE Int. Conf. Commun.*, 2020, pp. 1–6.
- [8] N. Guha, A. Talwalkar, and V. Smith, "One-shot federated learning," *CoRR*, vol. abs/1902.11175, 2019. [Online]. Available: <http://arxiv.org/abs/1902.11175>
- [9] M. Duan et al., "Astraea: Self-balancing federated learning for improving classification accuracy of mobile deep learning applications," in *Proc. IEEE 37th Int. Conf. Comput. Des.*, 2019, pp. 246–254.

- [10] N. Yoshida, T. Nishio, M. Morikura, K. Yamamoto, and R. Yonetani, "Hybrid-FL for wireless networks: Cooperative learning mechanism using non-IID data," in *Proc. IEEE Int. Conf. Commun.*, 2020, pp. 1–7.
- [11] V. Smith, C.-K. Chiang, M. Sanjabi, and A. S. Talwalkar, "Federated multi-task learning," in *Advances in Neural Information Processing Systems*, I. Guyon et al., Eds, vol. 30. San Jose, CA, USA: Curran Assoc., Inc., 2017.
- [12] S. Caldas, V. Smith, and A. Talwalkar, "Federated Kernelized multi-task learning," in *Proc. SysML Conf.*, 2018, pp. 1–3.
- [13] Y. Liu, Y. Kang, C. Xing, T. Chen, and Q. Yang, "A secure federated transfer learning framework," *IEEE Intell. Syst.*, vol. 35, no. 4, pp. 70–82, Jul./Aug. 2020.
- [14] J. Chen, J. Li, R. Huang, K. Yue, Z. Chen, and W. Li, "Federated transfer learning for bearing fault diagnosis with discrepancy-based weighted federated averaging," *IEEE Trans. Instrum. Meas.*, vol. 71, 2022, Art. no. 3514911.
- [15] F. Sattler, K.-R. Müller, and W. Samek, "Clustered federated learning: Model-agnostic distributed multitask optimization under privacy constraints," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 8, pp. 3710–3722, Aug. 2021.
- [16] A. Ghosh, J. Hong, D. Yin, and K. Ramchandran, "Robust federated learning in a heterogeneous environment," *CoRR*, vol. abs/1906.06629, 2019. [Online]. Available: <http://arxiv.org/abs/1906.06629>
- [17] C. Briggs, Z. Fan, and P. Andras, "Federated learning with hierarchical clustering of local updates to improve training on non-IID data," in *Proc. Int. Joint Conf. Neural Netw.*, 2020, pp. 1–9.
- [18] N. Shlezinger, S. Rini, and Y. C. Eldar, "The communication-aware clustered federated learning problem," in *Proc. IEEE Int. Symp. Inf. Theory*, 2020, pp. 2610–2615.
- [19] X. Ouyang, Z. Xie, J. Zhou, J. Huang, and G. Xing, "ClusterFL: A similarity-aware federated learning system for human activity recognition," in *Proc. 19th Annu. Int. Conf. Mobile Syst., Appl., Serv.*, 2021, pp. 54–66.
- [20] A. Ghosh, J. Chung, D. Yin, and K. Ramchandran, "An efficient framework for clustered federated learning," in *Proc. Advances in Neural Information Processing Systems*, H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, Eds, vol. 33. San Jose, CA, USA: Curran Assoc., Inc., 2020, pp. 19586–19597.
- [21] C. Chen, Z. Chen, Y. Zhou, and B. Kailkhura, "FedCluster: Boosting the convergence of federated learning via cluster-cycling," in *Proc. IEEE Int. Conf. Big Data*, 2020, pp. 5017–5026.
- [22] M. S. H. Abad, E. Ozfatura, D. GÜndüz, and O. Ercetin, "Hierarchical federated learning ACROSS heterogeneous cellular networks," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, 2020, pp. 8866–8870.
- [23] Z. Wang, H. Xu, J. Liu, Y. Xu, H. Huang, and Y. Zhao, "Accelerating federated learning with cluster construction and hierarchical aggregation," *IEEE Trans. Mobile Comput.*, vol. 22, no. 7, pp. 3805–3822, Jul. 2023.
- [24] C. Feng, H. H. Yang, D. Hu, Z. Zhao, T. Q. S. Quek, and G. Min, "Mobility-aware cluster federated learning in hierarchical wireless networks," *IEEE Trans. Wireless Commun.*, vol. 21, no. 10, pp. 8441–8458, Oct. 2022.
- [25] Z. Chen, D. Li, M. Zhao, S. Zhang, and J. Zhu, "Semi-federated learning," in *Proc. IEEE Wireless Commun. Netw. Conf.*, 2020, pp. 1–6.
- [26] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," in *Proc. Mach. Learn. Syst.*, 2020, vol. 2, pp. 429–450.
- [27] J. MacQueen, "Classification and analysis of multivariate observations," in *Proc. 5th Berkeley Symp. Math. Statist. Probab.*, 1967, pp. 281–297.
- [28] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proc. 2nd Int. Conf. Knowl. Discov. Data Mining*, 1996, pp. 226–231.
- [29] W. R. Heinzelman, A. Chandrakasan, and H. Balakrishnan, "Energy-efficient communication protocol for wireless microsensor networks," in *Proc. 33rd Annu. Hawaii Int. Conf. Syst. Sci.*, Maui, HI, USA, 2000, vol. 2, p. 10, doi: [10.1109/HICSS.2000.926982](https://doi.org/10.1109/HICSS.2000.926982).
- [30] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, Nov. 1998.
- [31] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. Int. Conf. Learn. Represent.*, 2015, pp. 730–734.
- [32] M. Hearst, S. Dumais, E. Osuna, J. Platt, and B. Scholkopf, "Support vector machines," *IEEE Intell. Syst. Appl.*, vol. 13, no. 4, pp. 18–28, Jul./Aug. 1998.
- [33] W. Shi, S. Zhou, Z. Niu, M. Jiang, and L. Geng, "Joint device scheduling and resource allocation for latency constrained wireless federated learning," *IEEE Trans. Wireless Commun.*, vol. 20, no. 1, pp. 453–467, Jan. 2021.
- [34] Y. Rubner, C. Tomasi, and L. J. Guibas, "The Earth mover's distance as a metric for image retrieval," *Int. J. Comput. Vis.*, vol. 40, no. 2, pp. 99–121, 2000.



Zhikun Chen received the B.S. degree in electronic information science and technology from Shandong University, Jinan, China, in 2018. He is currently working toward the Ph.D. degree with the College of Information Science and Technology, University of Science and Technology of China, Hefei, China.

His research interests include federated learning and wireless artificial intelligence.



Daofeng Li received the B.S. degree in electronic engineering and information science from the University of Science and Technology of China, Hefei, China, in 2018. He is currently working toward the M.E. degree with the College of Information Science and Technology, University of Science and Technology of China, Hefei, China.

His research interests include artificial intelligence and wireless communication.



Rui Ni (Member, IEEE) received the B.E. and Ph.D. degrees in electrical and electronics engineering from the University of Science and Technology of China, Hefei, China, in 2006 and 2011, respectively.

Since 2011, he has been the Principal Engineer with the Wireless Technology Laboratory, 2012 Laboratory, Huawei Tech. Co. Ltd., Shenzhen, China. His research involves various architectures of radio access network and core network from 2G to 5G, especially network slicing in 5G era. He has a rich wireless network related experience in engineering

practice. His current research interests include orbital angular momentum, large intelligence surface, and wireless big data processing for beyond 5G systems.



Jinkang Zhu (Life Member, IEEE) received the B.S. degree in electrical engineering from Sichuan University, Chengdu, China, in 1966.

In 1966, he joined the University of Science and Technology of China, Hefei, China, where he has been a Professor since 1992. He research interests include wireless mobile communications and networks, signal processing for communications, and future wireless technologies.



Sihai Zhang (Senior Member, IEEE) received the Ph.D. degree in computer science from the Department of Computer Science and Technology, University of Science and Technology of China (USTC), Hefei, China, in 2006.

He is currently an Associate Professor with the School of Micro-Electronic, USTC. He has authored or coauthored more than 60 technical papers. He initiated the research field of wireless big data in 2014. His research interests include wireless intelligence and wireless big data for communications and

networking, machine learning, and electronic design automation for integrated circuit design.